

A structured technical report covering message analysis methodologies for all supported channels.

1. **TASK 1**  study how message analysis is performed technically across:
 - a. Email
 - b. WhatsApp
 - c. Telegram
 - d. SMS

Message analysis means **examining communication data (metadata + content)** to understand who communicated, when, how often, and sometimes what was said. The technical approach differs depending on the platform because each uses different **protocols, encryption methods, and storage systems.**

1.2 Email Message Analysis

Email is one of the **easiest communication systems to analyze** because it uses open protocols.

Protocols Used

- **SMTP (Simple Mail Transfer Protocol)** → sending emails
- **POP3 (Post Office Protocol)** → downloading emails
- **IMAP (Internet Message Access Protocol)** → syncing mailboxes

What Analysts Examine

1. Email Headers

Headers contain routing information.

Example fields:

- **From**
- **To**
- **Subject**
- **Date**
- **Received**

- **Message-ID**
- **Return-Path**
- **DKIM signature**
- **SPF result**

Received: from mail.example.com

by gmail.com with ESMTP id...

This reveals:

- Sender mail server
- Relay servers
- IP addresses
- Timestamp chain

Tools Used

- **Wireshark**
- **Splunk**
- **Autopsy**

Types of Email Analysis

- Phishing detection
- Spam detection
- Sender IP tracing
- Email spoofing investigation

2 WhatsApp Message Analysis

WhatsApp uses strong encryption.

Encryption

WhatsApp uses the **Signal Protocol**.

This means:

- Messages are encrypted **on sender device**
- Decrypted **only on receiver device**
- Servers **cannot read messages**

What Can Be Analyzed

Since content is encrypted, analysts rely on:

1. Metadata

- sender phone number
- receiver phone number
- timestamp
- message size
- delivery status

2. Network Traffic

Using tools like:

- **Wireshark**

But packets appear as **TLS encrypted traffic**.

Client → WhatsApp server

Encrypted TLS packet

3. Device Forensics

Messages can be extracted from phone storage.

3 Telegram Message Analysis

Telegram uses a different architecture.

Encryption Types

1? Cloud Chats

- Encrypted **client** → **server**
- Stored on Telegram cloud
- Telegram can technically access data

2? Secret Chats

- End-to-end encrypted
- Messages stored **only on the devices**
- Not stored in Telegram cloud

What Analysts Can Analyze

Network Level

Packets between device and Telegram servers.

Tools:

- **Wireshark**

But payload remains encrypted.

Device Level

Telegram stores data locally in: cache4.db

4? SMS Message Analysis

SMS uses the **telecom signaling network**, not the internet.

Technology Used

SMS runs on:

- **SS7**: Call setup, SMS routing, Mobile roaming, Network signaling. ?contains vulnerabilities

or newer:

- Diameter protocol (used in 4G LTE)

Characteristics

✗ No encryption in traditional SMS

✓ Messages pass through **mobile carrier servers**

What Analysts Can Analyze

Metadata

- sender phone number
- receiver number
- timestamp
- SMS center

Network Data

Operators store **Call Detail Records (CDR)**.

Example: sender: +2010xxxx

Receiver: +2012xxxx

Time: 12:01

SMS Center: Vodafone Egypt

5.2 Technical Comparison

Platform	Encryptin	Content Visibility	Analysis Method
Email	Optional	Often visible	Header + server logs
WhatsApp	End-to-end	Not visible	Metadata + device forensics
Telegram	Mixed	Depends on chat type	Metadata + device data
SMS	None	Visible to carriers	Telecom network records

2. TASK 2 Email-Specific Research:

1 Email Header Structure:

Field	Purpose
From	Claimed sender address
To	Recipient address
Subject	Email title
Date	Time message was created
Message-ID	Unique email identifier
Reply-To	Address for replies
Return-Path	Bounce address
Received	Mail server routing path
DKIM-Signature	Cryptographic signature
AuthenticationResult	SPF / DKIM / DMARC results

2 SPF (Sender Policy Framework) Validation:

Sender Policy Framework is a DNS-based mechanism used to verify whether an email server is allowed to send mail for a domain.

How SPF Works

1  Domain owner publishes SPF record in DNS

2  Receiving server checks:

- sender domain
- sending IP

3  Result returned:

	Authorized
Pass	server
Fail	Unauthorized server

3? DKIM Validation: DomainKeys Identified Mail uses **cryptographic signatures** to verify email integrity.

How DKIM Works

1? Sending server signs the email using a **private key**

2? Signature added to header

3? Receiver retrieves **public key from DNS**

4? Receiver verifies the signature.

4? DMARC Policies: DMARC policies are the rules a domain owner publishes to tell receiving mail servers **what to do if an email fails authentication checks**.

They are part of **DMARC**, which works together with:

- **Sender Policy Framework (SPF)**
- **DomainKeys Identified Mail (DKIM)**

DMARC tells receiving servers **what to do if authentication fails**.

DMARC policy types:

POLICY	ACTION
NONE	monitor only
QUARANTINE	send to spam
REJECT	block message

5? Sender Spoofing Detection

Email spoofing occurs when attackers **forge the sender address**.

Detection techniques:

- ✓SPF failure
- ✓DKIM failure
- ✓Domain mismatch
- ✓Suspicious mail server

INDICATOR	EXPLANATION
MISMATCHED DOMAIN.	SENDER NOT ALIGNED WITH SPF.
NEW DOMAIN.	RECENTLY REGISTERED.
SUSPICIOUS IP.	KNOWN SPAM SERVER.

6? Reply-To Mismatch Analysis: Reply-To mismatch analysis is a technique used in email security and forensics to detect email spoofing or phishing. It checks whether the Reply-To address is different from the From address in the email header. If they are different, it may indicate a suspicious or malicious email.

Phishing emails often use different reply addresses

7? **Routing Chain Analysis:** Routing chain analysis means examining the path an email took from the sender to the receiver by analyzing the email headers.

Every mail server that handles the email adds a **“Received” header**, which creates a **chain of servers** the email passed through.

? Where the Routing Chain Is Found

You can see it in the **email header** under multiple lines like:

```
Received: from mail.example.com (192.168.1.5)
Received: from smtp.server.net (203.0.113.4)
Received: from unknown (45.23.12.90)
```

These lines show the **route of the email**.

Analysts Examine:

- originating IP
- mail relay servers
- geographic origin
- time differences

This helps detect:

- Botnets: A hacker infects computers (PCs, servers, IoT devices) with malware. These infected machines become “bots” (or zombies). All the bots connect to a central **command-and-control (C&C) server**, which the hacker uses to control them remotely.
- compromised servers
- phishing infrastructure

TASK 3 Define technical indicators that classify a message as suspicious:

1 Header Anomalies

Email headers contain routing and authentication information.

Header anomalies occur when fields contain abnormal or inconsistent values.

INDICATOR	EXPLANATION
MISSING AUTHENTICATION FIELD	No SPF/DKIM results present
SUSPICIOUS RECEIVED CHAIN	Email passed through unknown servers
MULTIPLE MESSAGE-ID FORMATS	Message-ID domain differs from sender
TIME INCONSISTENCIES UNUSUAL MAIL SERVERS	Received timestamps not sequential Mail originates from suspicious regions or hosts

Example

From: support@bank.com

Message-ID: [<12345@mail.ru>](mailto:12345@mail.ru)

The Message-ID domain (mail.ru) does not match bank.com, indicating possible spoofing.

Why This Matters

Attackers often use cheap or compromised mail servers that produce abnormal headers.

2□ Domain Mismatches

In **email security**, a **domain mismatch** means that **different parts of an email claim to be from different domains**, which is suspicious and often used in **phishing attacks**.

Fake email example

From: security@paypal.com

Reply-To: accountverify@gmail.com

Link: <http://paypal-login-security.xyz>

Here we have **three different domains**:

Field	Domain
From	paypal.com
Reply-To	gmail.com
Link	paypal-login-security.xyz

3□ Impersonation Patterns

Attackers frequently impersonate **trusted organizations or executives**.

These patterns are detected by analyzing **name formats, domains, and branding**.

Common Impersonation Techniques

Legitimate Domain	Fake Domain
Paypal.com	paypal.com
Microsoft.com	micr0soft.com
amazon.com	amaz0n-support.com

Attackers replace characters like:

- l → I
- o → 0
- rn → m

4▣ Metadata Inconsistencies

Metadata describes **how and when the message was transmitted**.

Inconsistencies in metadata often reveal spoofing or automation.

Common Metadata Issues

Indicator	Explanation
Timestamp gaps	Mail appears to travel faster than possible
IP location mismatch	Sender claims US company but originates from foreign IP
Bulk sending patterns	Same email sent to thousands of users
Abnormal sending time	Email sent at unusual hours

Example

Received: from company.com (New York)
Origin IP: 185.220.xxx.xxx (Russia)

This mismatch suggests the sender is **not actually the company**.

Phishing Evasion Techniques and Unresolved Challenges

1. Look-Alike Domain Attacks (Typosquatting)

Attackers register domain names that closely resemble legitimate domains in order to trick users.

Example:

Legitimate Domain	Malicious Domain
paypal.com	paypal.com
microsoft.com	micr0soft.com
amazon.com	amaz0n-support.com

These domains visually appear legitimate to users.

Challenge

Spam filters often rely on **blacklists or reputation systems**.

Newly registered domains may not yet appear in threat databases, allowing phishing emails to pass through security filters.

2. Compromised Legitimate Email Accounts

In some attacks, hackers gain access to **real corporate email accounts** and send phishing emails from them.

Example:

From: employee@company.com

Since the email is sent from a legitimate server:

- SPF passes
- DKIM passes
- DMARC passes

Challenge

Authentication systems verify **domain legitimacy**, but they cannot determine whether the sender's account has been compromised.

This type of attack is common in **Business Email Compromise (BEC)** incidents.

3. URL Obfuscation Techniques

Phishing emails frequently hide malicious links using obfuscation methods.

Examples:

Shortened links:

<https://bit.ly/3abX9K>

Deceptive subdomains:

<https://paypal.security-check.ru>

Challenge

Spam filters may fail to identify the final destination of shortened or complex URLs without performing deeper analysis.

4. HTML Content Manipulation

Attackers manipulate HTML elements within emails to hide malicious links.

Example:

Displayed link:

<https://paypal.com>

Actual hyperlink destination:

<https://attacker.ru>

Example HTML structure:

```
<a href="https://attacker.ru">https://paypal.com</a>
```

Challenge

Some spam filters analyze only visible text and may not fully inspect the underlying HTML code.

Why Phishing Detection Is Still Not Fully Resolved

Despite major improvements in email security technologies, phishing attacks remain a persistent problem. Several technical

and operational challenges prevent existing systems from completely eliminating phishing emails.

1. Legacy Email Infrastructure

Email protocols such as **SMTP** were designed decades ago without strong built-in security.

Key limitations include:

- No mandatory sender authentication in the original protocol
- Weak identity verification mechanisms
- Compatibility requirements with older mail servers

Because billions of systems rely on these protocols, **completely redesigning email infrastructure is extremely difficult.**

2. Incomplete Adoption of Security Standards

Technologies like SPF, DKIM, and DMARC are **optional** and not universally implemented.

Common issues include:

- Many domains do not publish DMARC policies
- Some organizations use weak configurations (e.g., p=none)
- Misconfigurations can break legitimate email delivery

As a result, attackers can exploit domains that lack proper authentication policies.

3. Rapid Domain and Infrastructure Creation

Attackers can create phishing infrastructure very quickly.

They can easily:

- register new domains
- create new email servers
- change hosting providers

These resources may exist for only **a few hours**, which is often faster than security systems can detect and blacklist them.

4. Human Behavior and Social Engineering

Phishing is not purely a technical problem; it also exploits **human psychology**.

Attackers use tactics such as:

- urgency (“verify your account immediately”)
- authority impersonation (CEO or bank emails)
- curiosity or fear

Even when technical indicators are weak, users may still trust the message.